

Tutorial 8 (solutions)

① Let $f: [a, b] \rightarrow \mathbb{R}$ be differentiable with $f(a) = 0$. If there is $M > 0$

s.t

$$|f'(x)| \leq M |f(x)|,$$

for every $x \in [a, b]$ then show that $f \equiv 0$.

Solution:

Given that f is differentiable and hence f is continuous.

Let $0 < \delta < \frac{1}{2M}$ be such that

$$|f(x) - f(y)| < 1 \quad \text{whenever} \quad |x - y| < \delta.$$

Now consider $x_0 \in (a, a+\delta)$.

By the mean value theorem

$$\exists x_1 \in (a, x_0) \text{ s.t.}$$

$$f(x_0) - f(a) = f'(x_1)(x_0 - a).$$

$$\Rightarrow |f(x_0)| = |f'(x_1)| |x_0 - a|$$

$$< M |f(x)| \delta$$

$$= \frac{|f(x)|}{2}.$$

In general $\exists x_n \in (a, x_{n-1})$

with

$$|f(x_{n-1})| < \frac{|f(x_n)|}{2}.$$

Hence $|f(x_0)| < \frac{|f(x_n)|}{2^n}$.

Since $f(a) = 0$ and

$$|f(x_n) - f(a)| < 1 \quad \text{we get}$$

$$|f(x_0)| < \frac{1}{2^n} \quad \text{for every } n \in \mathbb{N}$$

Thus $f \equiv 0$ in $[a, a+\delta]$.

Now repeat the argument with

$a+\delta$ in place of a and

we conclude that $f \equiv 0$ in $[a, b]$.

② Let $g: [-1, 1] \rightarrow [-1, 1]$ be differentiable surjective map with $|g'(x)| \leq 1$ for $-1 \leq x \leq 1$.

Then show that either

$$g(x) = x \quad (\text{or}) \quad g(x) = -x,$$

for every $x \in [-1, 1]$.

Solution: For every $x < y$, by mean value theorem $\exists u \in (x, y)$ s.t

$$\begin{aligned} |g(x) - g(y)| &= |g'(u)| |x - y| \\ &\leq |x - y|. \end{aligned}$$

Now we show that

$$g^{-1}(\{-1, 1\}) = \{-1, 1\}.$$

Since g is surjective

there exist $a, b \in [-1, 1]$

$$\text{s.t. } g(a) = -1, \quad g(b) = 1.$$

By the mean value theorem,

$$\exists c \in (a, b) \quad (\text{WLOG assume } a < b)$$

s.t.

$$g(b) - g(a) = g'(c)(b-a)$$

$$\Rightarrow |g(b) - g(a)| \leq |b-a|$$

$$\Rightarrow |1 - (-1)| \leq |b-a|$$

$$\therefore |b-a| \geq 2.$$

As $a, b \in [-1, 1]$ the only possible choice is

either $a = 1, b = -1$ (or)

$$a = -1, b = 1.$$

This shows that

$$g^{-1}(\{-1, 1\}) = \{-1, 1\}.$$

If $g(-1) = -1$ and $g(1) = 1$,

then by the mean value theorem

$\exists u_x \in (-1, x)$ s.t

$$g(x) - g(-1) = g'(u_x)(x+1)$$

$$\Rightarrow |g(x)+1| \leq |x+1|.$$

$$\Rightarrow -(x+1) \leq g(x)+1 \leq x+1$$

$$\text{So, } -x-2 \leq g(x) \leq x \quad \text{--- } \textcircled{\text{I}}$$

Also, by the mean value theorem,

$$\exists \, v_x \in (x, 1) \text{ s.t.}$$

$$g(1) - g(x) = g'(v_x)(1-x)$$

$$\Rightarrow |1 - g(x)| \leq |1-x|$$

$$\Rightarrow -(1-x) \leq 1 - g(x) \leq 1-x$$

so,

$$x \leq g(x) \leq 2-x$$

————— $\textcircled{\text{II}}$

From $\textcircled{\text{I}}$ and $\textcircled{\text{II}}$ we get

$$\boxed{g(x) = x}$$

Now consider the case

$$g(-1) = 1 \quad \text{and} \quad g(1) = -1$$

and follow the similar steps, we get

$$\boxed{g(x) = -x}$$

③ Give an example of a uniformly continuous on $[0,1]$ that is differentiable on $(0,1)$ but whose derivative is not bounded on $(0,1)$.

Solution:

Take $f(x) = \sqrt{x}$ on $[0,1]$

Here f is uniformly continuous and it is differentiable on $(0,1)$. However,

$$f'(x) = \frac{1}{2\sqrt{x}} \quad \text{is}$$

NOT bounded on $(0,1)$.

④ Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$, $\forall x \geq 0$. Show that $f(x) \leq g(x)$, for all $x \geq 0$.

Solution:

Since f, g are differentiable on $\mathbb{R}$, it follows that $(g-f)$ is

differentiable on $\mathbb{R}$.

$$\begin{aligned}\text{Clearly } (g-f)(0) \\ &= g(0) - f(0) \\ &= 0.\end{aligned}$$

For $x > 0$, by applying the mean value theorem to $g-f$ on $[0, x]$, we see that $\exists c \in (0, x)$ s.t

$$\begin{aligned}(g-f)(x) - (g-f)(0) \\ &= (g-f)'(c)(x-0)\end{aligned}$$

$$\Rightarrow g(x) - f(x) = [g'(c) - f'(c)]x$$

Since $c > 0$ & $f'(c) \leq g'(c)$

we have

$$g(x) - f(x) \geq 0.$$

Thus $f(x) \leq g(x)$,

for any $x \geq 0$.

⑤ If a rock is dropped from a height of 150 feet, its position t seconds after it is dropped until it hits

the ground is given by
the function $s(t) = -t^2 + 5t + 150$

② Determine how long it
takes before the rock
hits the ground.

Solution:

The rock is dropped
from 150 feet height, the
position after t seconds
is given by (position function)

$$s(t) = -t^2 + 5t + 150.$$

Hitting the ground means

$$s(t) = 0$$

$$\Leftrightarrow t^2 - 5t - 150 = 0.$$

$$t = \frac{5 \pm \sqrt{25 + 600}}{2}$$

$$= \frac{5 \pm 25}{2}$$

$$= 15 \text{ (or) } -10$$

$$\Rightarrow \boxed{t = 15 \text{ sec}}$$

The rock hits the ground
after 15 seconds.

⑥ Find the average velocity V_{avg} of the rock when it is released and hits the ground.

Solution:

The average velocity V_{avg} is given by

$$V_{avg} = \frac{s(15) - s(0)}{15}$$

$$= \frac{-150}{15}$$

$$= -10 \text{ ft/sec.}$$

② Find the time t when the instantaneous velocity of the rock is V_{avg} .

Solution: The instantaneous velocity is given by the derivative of position function

Thus we need to find a time ' t ' such that

$$v(t) = s'(t) = -10 \text{ ft/sec}$$

Since $s(t)$ is continuous

on the interval $[0, 15]$

and differentiable over

$(0, 15)$, by the mean

value theorem $\exists c \in (0, 15)$

such that

$$s'(c) = \frac{s(15) - s(0)}{15}$$

$$= -10$$

Since $s'(t) = -2t + 5$,

we have

$$-2C + 5 = -10$$

$$\Rightarrow \boxed{C = 7.5}$$

Therefore, after 7.5 sec
(of dropping rock) the
instantaneous velocity
equals the average velocity
of the rock during its
free fall. 

